



mob. no.- 8077417849
email- anshuljain01571@gmail.com
linkedin.com/in/anshul-jain-8b2902216

ANSHUL JAIN



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
2026	M.Tech in Optoelectronics & Optical Communication	Indian Institute of Technology Delhi	7.47
2019	B.Sc.(Hons.)	ALIGARH MUSLIM UNIVERSITY	7.61
2014	RBSE	SHREE SK DEORA SCHOOL	79.40
2012	RBSE	SAINT THOM ACADEMY	81.83
2021	M.Sc.	ALIGARH MUSLIM UNIVERSITY	7.88

SCHOLASTIC ACHIEVEMENTS

- Joint CSIR-UGC(NET/JRF) June-Dec 2023, AIR:217
- Graduate Aptitude Test in Engineering (GATE)2024, AIR:431

IIT DELHI THESIS

Title: Beamforming in optical wireless communications

Supervisor: Prof. Swades Kumar De, Communication Networks Research Group Lab., IIT Delhi

Description:- Signal Processing and Machine Learning

- Developed **CNN**-based WFS-less wavefront correction achieving **0.0819 rad RMSE** for dynamic atmospheric turbulence, predicting 15 Zernike coefficients from PSF images in real-time.
- Studying Algorithms(**SPGD** , **SA**, **GA** etc.) based on **DNN** and **ML** for correcting wavefront aberration.

PROJECTS

- **Domain-Specific RAG System with Hybrid Retrieval & Re-ranking** (Apr, 2026 - May, 2026):
 - Built a production-grade Retrieval-Augmented Generation (**RAG**) assistant over **FastAPI**
 - Engineered full data pipeline: sitemap-driven scraper → two-stage chunking (markdown-header split + token-aware fallback via *tiktoken*) → metadata-rich JSONL storage with source URL & heading breadcrumbs.
 - Implemented **hybrid retrieval** combining dense semantic search (*BAAI/bge-small-en-v1.5* embeddings on **ChromaDB**) with **BM25** keyword search, fused via **Reciprocal Rank Fusion (RRF)**
 - Added *BAAI/bge-reranker-base* **cross-encoder re-ranking** (retrieve top-20 → rerank → return top-5)
 - Evaluated end-to-end with **RAGAS** on an 18-question hand-curated eval set spanning single-library and cross-library queries.
 - **Stack:** Python, sentence-transformers, ChromaDB, rank-bm25, tiktoken, Anthropic API (Claude Haiku), RAGAS, Streamlit.

- **CLASSIFYING PERSONAL INCOME (Prof. Ragunathan Rengasamy)** (Jan, 2025 - Feb, 2025) :(IIT Madras)
 - Case Study on **Classification** learn how to preprocess the dataset, explore data, **Cross-tabulation** and **Data- Visualisation**
 - **Logistic Regression**, test & train data, Predict an individual's income bracket.
 - concept of dummy variables for incorporating categorical variables into the model.
 - evaluate the model's performance using metrics like accuracy and refining model.

[GitHub Link](#)

TECHNICAL SKILLS

- **Programming and other Languages:** Python | C++ | SQL
- **Packages and Libraries:** Numpy, Pandas, Matplotlib, Seaborn, OOPAO, Scikit-learn
- **Software and Tools:** Jupyter Notebook, IDLE, VS Code, My Office(Word, Excel, PowerPoint), GitHub

- **NPTEL Courses** : Python For **Data Science**, IIT Madras (Prof. Raghunathan Rengasamy)
 - **Artificial Intelligence- Search Method for Problem Solving**, IIT Madras (Prof. Deepak Khemani)
- **Coursera:** Microsoft Azure Machine Learning for **Data Scientists**

AREAS OF INTEREST

- Deep Learning with a focus on Computer Vision for real-time applications.
- Developing predictive models for customer churn and lifetime value.
- Exploring Generative AI for creative and data augmentation tasks.

TEACHING ASSISTANTSHIP

- Assisted Students in JOP **Fiber Optics Lab.** with conducting experiments in Laser, **WDM,OTDR,Optical Fibres, EDFA** and **Splicing Machine**.



ANSHUL JAIN



IIT COURSE

Degree	Institute	CGPA	Dept. Rank
M.Tech in Optoelectronics & Optical Communication	Indian Institute of Technology Delhi	7.47	---

COURSES DONE

Fiber Optics, Fiber Optics And Optical Communications Lab., Photonic Devices, Digital Communication & Inform, Optoelectronic Instrumentation, Optical Electronics, Biomedical Optics & Bio-photo., Optical Communication Systems, Micro And Nano Photonics,Optics and Laser, Fiber Optics And Optical Communication Lab.ii

SUMMARY

M.Tech graduate from IIT Delhi with research in ML-based signal processing, transitioning into applied AI engineering. Built RAG systems and LLM-powered applications using LangChain, FastAPI and vector databases. Strong foundation in Python, statistical modelling and end-to-end ML pipelines.